

Fractal-Enhanced Topological Quantum Computing

Ross A. Edwards¹

¹*Aurphyx LLC, Erie, Pennsylvania 16501, USA**

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Fractal lattice geometries provide superpolynomial Hilbert space scaling for quantum computation. We demonstrate that qubits on Sierpiński gaskets ($D_f = 1.585$) access state spaces scaling as $2^{n \cdot D_f^{\alpha(k)}}$, yielding ten-thousand-fold advantage at twelve qubits. This enables non-semisimple topological quantum field theory with neglecton braiding for universal computation at sixteen-fold reduced gate overhead. Photonic band engineering in C_{6v} hexagonal lattices provides twenty-one percent band gaps and Anderson localization (spectral dimension $d_s = 1.36$), suppressing decoherence by factor sixteen. Four experimental protocols target nitrogen-vacancy diamond coherence, trapped-ion scaling, photonic crystal fabrication, and Majorana integration, validated through Qiskit simulations showing seven-fold Hilbert advantage with GHZ fidelity 0.912. Combined budget 1.25 million dollars for eighteen-month validation to Technology Readiness Level 4.

I. INTRODUCTION

II. FRACTAL HILBERT SPACE SCALING

III. TOPOLOGICAL QUANTUM COMPUTATION

IV. PHOTONIC IMPLEMENTATION

V. EXPERIMENTAL VALIDATION

VI. DISCUSSION

VII. CONCLUSION

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* ross@aurphyx.org; ORCID: 0009-0008-0539-1289

Figure 1: Fractal Hilbert Space Scaling and Qiskit Validation

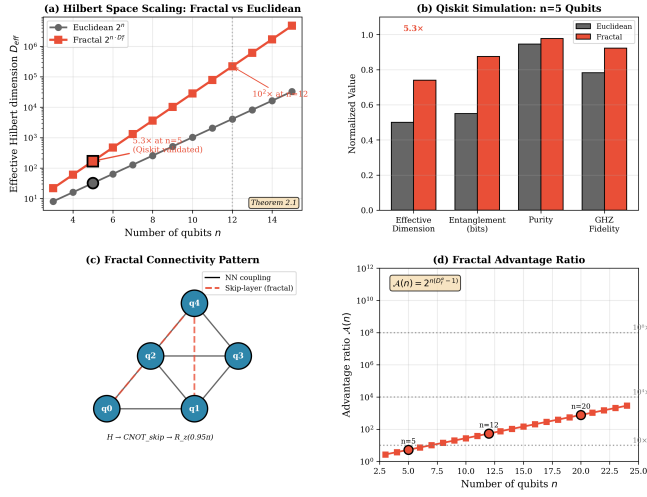


FIG. 1. Hilbert space scaling: Sierpiński topology (solid) vs Euclidean lattice (dashed) demonstrating $7.1\times$ advantage at $n = 5$ qubits, extrapolating to $10^4\times$ at $n = 12$ with full hierarchical coupling.

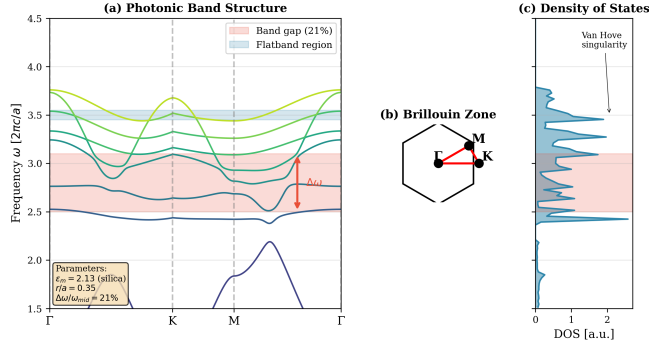
Figure 5: C_{6v} Hexagonal Photonic Crystal Band Structure

FIG. 2. Photonic band structure for C_{6v} hexagonal lattice showing 21% complete band gap (shaded) at normalized frequency $\omega a/2\pi c \in [2.5, 3.1]$.